Practice workbook · Use ABO antigens and antibodies to choose donors and explain clumping

Antigens and antibodies of each blood group, which donors suit which recipients, counting compatible people, and why mismatches cause agglutination.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-ABO-ANTIGENS-AND-TRANSFUSION

1. Question 4 · part c · 4 marks

November 2020 Paper 1

Worked-example retry · Attempt part c only.

4 There are four different blood groups, A, B, AB and O.

The antigens on the surface of red blood cells determine a person's blood group. There are two types of antigen, antigen A and antigen B.

(a) The table lists the four different blood groups.

Complete the table by giving the correct type of antigens for each blood group.

(2)

Blood group	A	B	AB	O
Antigen				

(b) If a person needs a blood transfusion, it is important that they receive a suitable blood group.

Explain the effects on a person who receives a blood group that is not suitable.

(3)

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1. Question 4 · part c · 4 marks

November 2020 Paper 1

Worked-example retry · Attempt part c only.

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(c) Explain which blood group or groups can be given to people with each of the other blood groups during a transfusion.

(4)

(d) The table shows the percentage of students with each type of blood group in a school.

Blood group	Percentage (%)
A	42
B	10
AB	4
O	44

There are 750 students in the school.

Calculate the number of students who can safely receive blood group B.

(3)

number of students = ...

(Total for Question 4 = 12 marks)

1. Question 4 · part c · 4 marks · Working and answer

Working and answer

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2. Question 4 · part a · 2 marks

November 2020 Paper 1

Further practice · Attempt part a only.

4 There are four different blood groups, A, B, AB and O.

The antigens on the surface of red blood cells determine a person's blood group. There are two types of antigen, antigen A and antigen B.

(a) The table lists the four different blood groups.

Complete the table by giving the correct type of antigens for each blood group.

(2)

Blood group	A	B	AB	O
Antigen				

(b) If a person needs a blood transfusion, it is important that they receive a suitable blood group.

Explain the effects on a person who receives a blood group that is not suitable.

(3)

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2. Question 4 · part a · 2 marks · Working and answer

Working and answer

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3. Question 4 · part d · 3 marks

November 2020 Paper 1

Further practice · Attempt part d only.

4 There are four different blood groups, A, B, AB and O.

The antigens on the surface of red blood cells determine a person's blood group. There are two types of antigen, antigen A and antigen B.

(a) The table lists the four different blood groups.

Complete the table by giving the correct type of antigens for each blood group.

(2)

Blood group	A	B	AB	O
Antigen				

(b) If a person needs a blood transfusion, it is important that they receive a suitable blood group.

Explain the effects on a person who receives a blood group that is not suitable.

(3)

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3. Question 4 · part d · 3 marks

November 2020 Paper 1

Further practice · Attempt part d only.

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(c) Explain which blood group or groups can be given to people with each of the other blood groups during a transfusion.

(4)

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(d) The table shows the percentage of students with each type of blood group in a school.

Blood group	Percentage (%)
A	42
B	10
AB	4
O	44

There are 750 students in the school.

Calculate the number of students who can safely receive blood group B.

(3)

number of students = ...

(Total for Question 4 = 12 marks)

3. Question 4 · part d · 3 marks · Working and answer

Working and answer

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4. Question 9 · part a.i · 2 marks

June 2024 Paper 1

Further practice · Attempt part a.i only.

9 (a) There are four main blood groups.

One of these groups is known as type O.

(i) Name the other three blood groups.

(2)

(ii) A person with blood type O needs a blood transfusion.

Explain which blood type this person should receive in a transfusion.

(5)

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4. Question 9 · part a.i · 2 marks · Working and answer

Working and answer

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5. Question 4 · part b · 3 marks

November 2020 Paper 1

Further practice · Attempt part b only.

4 There are four different blood groups, A, B, AB and O.

The antigens on the surface of red blood cells determine a person's blood group. There are two types of antigen, antigen A and antigen B.

(a) The table lists the four different blood groups.

Complete the table by giving the correct type of antigens for each blood group.

(2)

Blood group	A	B	AB	O
Antigen				

(b) If a person needs a blood transfusion, it is important that they receive a suitable blood group.

Explain the effects on a person who receives a blood group that is not suitable.

(3)

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5. Question 4 · part b · 3 marks

November 2020 Paper 1

Further practice · Attempt part b only.

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(c) Explain which blood group or groups can be given to people with each of the other blood groups during a transfusion.

(4)

(d) The table shows the percentage of students with each type of blood group in a school.

Blood group	Percentage (%)
A	42
B	10
AB	4
O	44

There are 750 students in the school.

Calculate the number of students who can safely receive blood group B.

(3)

number of students = ...

(Total for Question 4 = 12 marks)

5. Question 4 · part b · 3 marks · Working and answer

Working and answer

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6. Question 2 · part a.iii · 3 marks

November 2020 Paper 2R

Further practice · Attempt part a.iii only.

2 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Treatment of Ebola

Ebola is caused by an RNA virus. There is no proven treatment available for Ebola. However, whole blood collected from someone who has recovered from infection has been used as a treatment. The results have been promising in a small group of Ebola patients.

Guidance for national health authorities and blood transfusion services describes the stages needed to collect whole blood (CWB) or plasma (CP) from Ebola-recovered patients. This can be used for transfusion to patients as a treatment for early Ebola.

The guidance states how to

- identify people who have recovered from Ebola as potential blood donors
- screen donors' blood for A, B, O and Rhesus groupings
- screen donors' blood for transfusion-transmissible infections
- collect blood and care for donors
- obtain agreement of Ebola patients for the treatment
- identify patient's blood grouping
- store and transport CWB and CP to the places where transfusions are to be given
- select Ebola patients for transfusion
- transfuse the blood into the patient

Donated CWB should never be frozen and should be stored between 2 °C and 6 °C.

CWB and CP donations should be transfused to the Ebola patients using standard transfusion procedures.

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6. Question 2 · part a.iii · 3 marks

November 2020 Paper 2R

Further practice · Attempt part a.iii only.

(a) (i) Explain what is meant by an RNA virus (line 1).

(2)

(ii) The table lists five diseases. Put a tick (✓) next to the diseases that should be tested for in the donors' blood.

(2)

Disease	Blood tested (✓)
anaemia	
cystic fibrosis	
gonorrhoea	
HIV	
scurvy	

(iii) Explain why it is necessary to identify the blood group of the blood collected from donors who have recovered from Ebola (line 9).

(3)

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6. Question 2 · part a.iii · 3 marks

November 2020 Paper 2R

Further practice · Attempt part a.iii only.

(iv) Suggest why donated CWB should never be frozen (line 17).

(2)

(v) Explain how the blood of people who have recovered from Ebola is likely to be effective in treating the disease.

(2)

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(b) The nitrogenous base composition of another virus was found to be adenine 13%, guanine 26%, uracil 20% and cytosine 41%.

Explain how these figures prove that this is a virus with a single strand of RNA.

(3)

(Total for Question 2 = 14 marks)

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6. Question 2 · part a.iii · 3 marks · Working and answer

Working and answer

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7. Question 4 · part a.ii · 3 marks

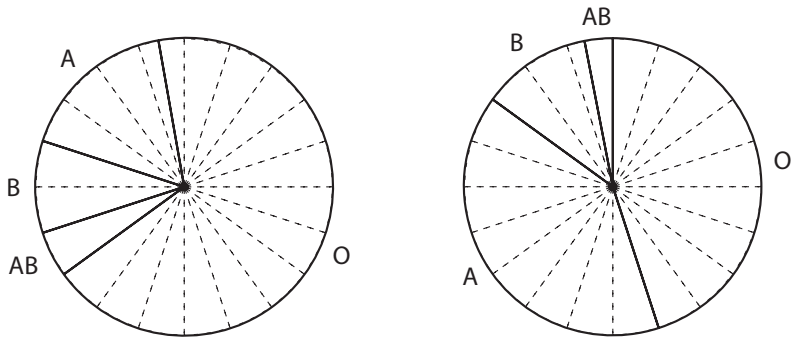
June 2022 Paper 1

Further practice · Attempt part a.ii only.

4 (a) Humans belong to one of four blood groups.

These blood groups are A, B, AB and O.

The pie charts show estimates of the percentages of people from Africa and the percentages of people from Europe in each blood group.



(i) Complete the tables to give the percentage of each blood group.

Some information has been given.

(2)

Africa

Blood group	Percentage of blood group
O	68
A	
B	
AB	5

Europe

Blood group	Percentage of blood group
O	45
A	
B	12
AB	

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P 7 0 9 7 3 A 0 1 2 2 4

7. Question 4 · part a.ii · 3 marks

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Further practice · Attempt part a.ii only.

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(ii) Some people need a blood transfusion.

Explain why people with blood group A cannot receive a blood transfusion from people with blood group B.

(3)

(iii) A male with the genotype I^AI^O and a female with the genotype I^BI^O have a child with the genotype I^OI^O.

Draw a genetic diagram to show how the child inherits the genotype I^OI^O.

(3)

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(iv) Determine the probability that a child from these parents will inherit the blood group AB.

(1)

probability = ...



7. Question 4 · part a.ii · 3 marks · Working and answer

Working and answer

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8. Question 9 · part a.ii · 5 marks

June 2024 Paper 1

Further practice · Attempt part a.ii only.

9 (a) There are four main blood groups.

One of these groups is known as type O.

(i) Name the other three blood groups.

(2)

(ii) A person with blood type O needs a blood transfusion.

Explain which blood type this person should receive in a transfusion.

(5)

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8. Question 9 · part a.ii · 5 marks

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Further practice · Attempt part a.ii only.

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(b) Haemophilia is a sex-linked genetic condition that affects the blood.

(i) Describe what is meant by a sex-linked genetic condition.

(2)

(ii) State one effect that haemophilia will have on an individual.

(1)

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(Total for Question 9 = 10 marks)

8. Question 9 · part a.ii · 5 marks · Working and answer

Working and answer

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9. Question 9 · part b.i · 3 marks

June 2019 Paper 1

Further practice · Attempt part b.i only.

9 Inheritance of blood group involves codominant inheritance.

(a) (i) Which statement describes codominant inheritance in ABO blood groups?

(1)

- ☒ **A** the inheritance of two different alleles, both of which are expressed
- ☒ **B** the inheritance of two different alleles, only one of which is expressed
- ☒ **C** the inheritance of multiple alleles, only two of which are expressed
- ☒ **D** the inheritance of multiple alleles, only one of which is expressed

(ii) State the possible genotypes of a person with blood group A.

(1)

(b) (i) A person's blood group is determined by antigens.

These antigens are carbohydrate and protein molecules on the surface of red blood cells.

In 2007, a team of scientists used enzymes to convert blood groups A, B and AB into blood group O for transfusions.

Suggest how enzymes can convert blood groups A, B and AB into blood group O.

(3)

(ii) Suggest an advantage of producing blood group O using enzymes, compared with other methods of obtaining blood group O.

(1)

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9. Question 9 · part b.i · 3 marks · Working and answer

Working and answer

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10. Question 9 · part b.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.ii only.

9 Inheritance of blood group involves codominant inheritance.

(a) (i) Which statement describes codominant inheritance in ABO blood groups?

(1)

- ☒ **A** the inheritance of two different alleles, both of which are expressed
- ☒ **B** the inheritance of two different alleles, only one of which is expressed
- ☒ **C** the inheritance of multiple alleles, only two of which are expressed
- ☒ **D** the inheritance of multiple alleles, only one of which is expressed

(ii) State the possible genotypes of a person with blood group A.

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These antigens are carbohydrate and protein molecules on the surface of red blood cells.

In 2007, a team of scientists used enzymes to convert blood groups A, B and AB into blood group O for transfusions.

Suggest how enzymes can convert blood groups A, B and AB into blood group O.

(3)

(ii) Suggest an advantage of producing blood group O using enzymes, compared with other methods of obtaining blood group O.

(1)

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10. Question 9 · part b.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.ii only.

(c) Haemophilia is a sex-linked blood disorder that reduces the ability of the blood to clot.

These are the genotypes of four offspring, P, Q, R and S.

P	Q	R	S
X^HX^h	X^hY	X^HX^H	X^HY

(i) Draw a genetic diagram to show how these offspring are produced from one set of parents.

(2)

(ii) These parents are expecting another baby.

Determine the probability that this baby will have haemophilia.

(1)

probability = ...

(Total for Question 9 = 9 marks)

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10. Question 9 · part b.ii · 1 mark · Working and answer

Working and answer

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